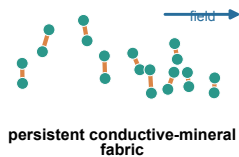
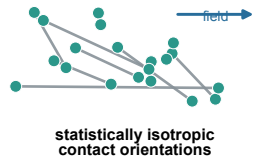


a

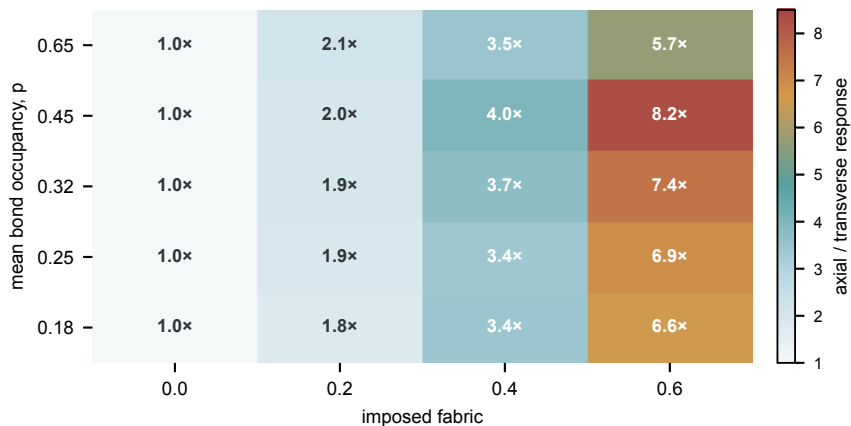
Two idealized end-members of contact-orientation order



random orientations
self-average with increasing volume

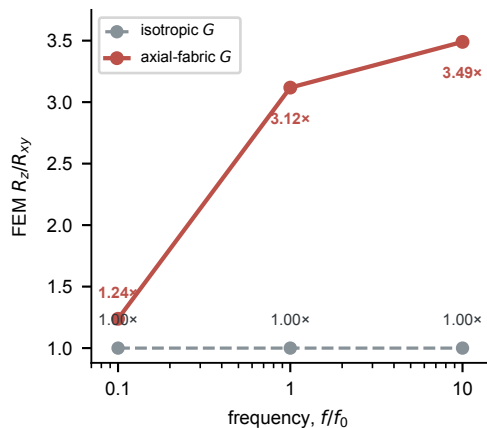
persistent fabric
directional response remains visible

b

Persistent fabric restores R_z/R_{xy} at $N = 1728$ 

At $\bar{\omega} = 1$: fabric 0.2, 0.4 and 0.6 produce R_z/R_{xy} of 1.8–2.1, 3.4–4.0 and 5.7–8.2.

Full-FEM validation
c same geometry, graph and total conductance



Fine-mesh ratio change at $f/f_0 = 1$: 0.26%.